# Artificial Intelligence for Cyber Defense: A Predictive Analytics and Risk-Based Decision Framework Using UNSW-NB15

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***Abstract*—**Cybersecurity analytics systems are valuable only when predictive outputs can be translated into timely and defensible actions. This study presents and evaluates an AI-driven predictive analytics pipeline that connects binary network-threat detection with a simple strategic decision-intelligence layer. A prepared subset of the UNSW-NB15 benchmark was used, containing 50,000 training records and 20,000 testing records. Forty-two predictive attributes were retained after excluding the binary label and attack-category metadata. Four classifiers—Logistic Regression, Random Forest, XGBoost, and a multilayer perceptron (MLP)—were trained under a common preprocessing pipeline and evaluated using accuracy, precision, recall, F1-score, ROC-AUC, training time, and inference time. XGBoost produced the strongest overall F1-score (0.8955), with 0.8733 accuracy, 0.8202 precision, 0.9861 recall, and 0.9841 ROC-AUC. Random Forest achieved the highest attack recall (0.9886), while XGBoost provided a more favorable overall performance-efficiency balance. Permutation importance identified source time-to-live (sttl) as the dominant feature in the fitted XGBoost pipeline. Finally, attack probabilities were mapped to experimental risk levels and recommended actions, illustrating how predictive analytics can be extended beyond classification toward operational decision support. The findings support tree-based ensemble methods for structured intrusion-detection data while also exposing the need to manage false positives and validate risk thresholds before operational deployment.

***Index Terms*—**cybersecurity, intrusion detection, predictive analytics, machine learning, XGBoost, UNSW-NB15, decision intelligence, risk scoring.

## I. Introduction

Modern network defense increasingly depends on the ability to convert high-volume telemetry into timely security judgments. Machine-learning-based intrusion detection provides a data-driven mechanism for identifying malicious behavior that may not be captured by static signatures. Cybersecurity data science therefore combines security data, analytical methods, and machine learning to support more intelligent protection and decision making [1]. At the same time, benchmark quality remains central to credible evaluation because intrusion-detection results are strongly affected by traffic composition, labels, collection environment, and attack coverage [2].

The UNSW-NB15 dataset was created to provide a more contemporary benchmark containing normal traffic and nine attack families, including Fuzzers, Analysis, Backdoors, denial-of-service (DoS), Exploits, Generic attacks, Reconnaissance, Shellcode, and Worms [3]. The official corpus includes millions of records and designated training and testing partitions. This study uses a reproducibly prepared local subset of those partitions to evaluate four widely used classification approaches under one consistent pipeline.

A second challenge is that a high-performing classifier does not by itself constitute decision support. Security operations require prioritization: which alerts deserve immediate attention, which can be monitored, and what action should follow. This motivates the concept of strategic decision intelligence used in this work: predictive attack probabilities are converted into explicit risk bands and corresponding recommended responses. The mapping is intentionally treated as an experimental design layer rather than a universal security standard.

The study addresses the following research question: How effectively can AI-driven predictive analytics detect cybersecurity threats and transform model predictions into explainable, risk-aware strategic decision intelligence? The contributions are fourfold: (1) a common local pipeline comparing linear, ensemble, boosting, and neural-network classifiers; (2) an evaluation emphasizing F1-score, attack recall, ROC-AUC, and computational time; (3) model-agnostic permutation importance for the best classifier; and (4) a probability-to-risk decision layer that connects detection output to recommended operational actions.

## II. Related Work

Intrusion detection has long been framed as a classification and anomaly-detection problem. Sarker et al. describe cybersecurity data science as the use of machine learning and data-driven modeling to extract patterns from security data and support intelligent decision making [1]. Ring et al. emphasize that the selection of network intrusion datasets is itself a methodological choice and propose criteria for assessing dataset suitability, including data volume and recording environment [2]. These observations justify both the use of a recognized benchmark and explicit reporting of the experimental partition used in this study.

Moustafa and Slay introduced UNSW-NB15 as a hybrid benchmark combining modern normal activity with synthesized contemporary attack behavior [3]. Subsequent work has used machine learning with UNSW-NB15 for network-forensic and botnet-related analysis, demonstrating the continued relevance of flow-oriented features for security classification [4]. Artificial neural networks have also shown potential for distinguishing malicious and benign traffic, although performance depends strongly on dataset, architecture, and evaluation design [5].

Among ensemble methods, XGBoost implements scalable gradient-boosted decision trees and includes algorithmic and systems optimizations for efficient tree learning [6]. Its suitability for structured tabular features

makes it a natural comparator to Random Forest, Logistic Regression, and an MLP in this study. For explainability, the broader XAI literature stresses that complex predictive performance must be accompanied by mechanisms that make model behavior understandable to relevant users [7]. SHAP is one influential explanation framework [8]; however, the present implementation uses permutation importance because it is model-agnostic and directly measures the reduction in held-out F1-score when a feature is disrupted.

## III. Dataset and Preprocessing

### *A. Dataset*

The experiment uses a prepared subset of the official UNSW-NB15 training and testing partitions. The local training set contains 50,000 records and the test set contains 20,000 records. Each file has 44 columns: 42 predictive features, the original attack-category field, and the binary label. The binary target is defined as 0 for Normal and 1 for Attack. The attack-category field is retained only as test metadata and is excluded from model inputs to prevent direct leakage of the class semantics.

TABLE I. Binary Class Distribution of the Prepared Local Dataset.

| Split | Normal | Attack | Total |
|---|---|---|---|
| Training | 15,969 | 34,031 | 50,000 |
| Testing | 8,988 | 11,012 | 20,000 |

All nine UNSW-NB15 attack families remain represented in the local data. In the test set, the largest attack groups are Generic (4,584), Exploits (2,704), Fuzzers (1,473), DoS (993), and Reconnaissance (849); smaller categories include Analysis, Backdoor, Shellcode, and Worms. This composition is relevant when interpreting aggregate binary metrics because performance is dominated by the more frequent classes.

### *B. Feature Processing*

The pipeline treats proto, service, and state as categorical attributes. Missing categorical values are imputed with the most frequent value and then one-hot encoded with unseen-category handling enabled. All remaining predictive variables are treated as numerical; missing values are median-imputed and standardized. The same preprocessing design is embedded inside each scikit-learn Pipeline, reducing the risk of inconsistent transformations across models. The local prepared files contained no missing or infinite values, but imputation was retained as a defensive pipeline step.

## IV. Proposed Predictive Decision-Intelligence Framework

The proposed workflow consists of five linked stages: (1) network-flow data acquisition, (2) preprocessing and feature transformation, (3) attack-probability estimation, (4) model interpretation and risk categorization, and (5) recommended strategic action. Unlike a classifier-only design, the final stage explicitly converts model confidence into a form intended for prioritization by security personnel.

TABLE II. Predictive Analytics to Strategic Decision-Intelligence Workflow.

| Stage | Function | Output |
|---|---|---|
| Data | UNSW-NB15 flow records and labels | Structured security observations |
| Analytics | Imputation, scaling, one-hot encoding | Model-ready features |
| AI prediction | LR, RF, XGBoost, MLP | Binary prediction and attack probability |
| Interpretation | Permutation importance | Ranked influential features |
| Decision intelligence | Probability thresholds and action mapping | Risk level and recommended response |

Risk levels are assigned using four experimental thresholds: Low for $p < 0.25$, Moderate for $0.25 \le p < 0.50$, High for $0.50 \le p < 0.75$, and Critical for $p \ge 0.75$. Recommended actions range from continued monitoring at Low risk to immediate incident-response review and containment procedures at Critical risk. These thresholds are research-design choices and require calibration to organizational costs, tolerance for false alarms, and incident-response capacity before production use.

## V. Experimental Methodology

### *A. Models*

Four binary classifiers were evaluated. Logistic Regression used balanced class weights and a maximum of 1,000 iterations. Random Forest used 200 trees, unrestricted depth, balanced-subsample class weighting, and random seed 42. XGBoost used 250 estimators, maximum depth 6, learning rate 0.08, subsample ratio 0.9, column-sampling ratio 0.9, and log-loss evaluation. The MLP used two hidden layers of 64 and 32 neurons, ReLU activation, batch size 256, initial learning rate 0.001, early stopping, a 15% internal validation fraction, and maximum 60 iterations. Random seed 42 was used where supported.

### *B. Evaluation Metrics*

The positive class is Attack. Performance is evaluated with accuracy, precision, recall, F1-score, and ROC-AUC. Because the security cost of missed attacks can be high and the test set is not class-balanced, recall and F1-score are especially informative. F1-score was selected as the primary model-ranking metric in the implementation. Wall-clock model training and inference times were also recorded to provide a basic efficiency comparison. No statistical significance testing or repeated cross-validation was performed; consequently, the reported timing and point estimates should be interpreted as results from this specific run and environment.

## VI. Results

TABLE III. Binary Threat-Detection Performance on 20,000 Held-Out Test Records.

| Model | Acc. | Prec. | Recall | F1 | AUC |
|---|---|---|---|---|---|
| XGBoost | 0.8733 | 0.8202 | 0.9861 | 0.8955 | 0.9841 |
| Random Forest | 0.8670 | 0.8112 | 0.9886 | 0.8912 | 0.9805 |
| MLP Neural Network | 0.8551 | 0.8056 | 0.9712 | 0.8807 | 0.9711 |
| Logistic Regression | 0.8387 | 0.8029 | 0.9371 | 0.8648 | 0.9575 |

XGBoost ranked first by F1-score, achieving 0.8733 accuracy, 0.8202 attack precision, 0.9861 attack recall, 0.8955 F1-score, and 0.9841 ROC-AUC. Random Forest ranked second by F1-score (0.8912) and achieved the highest attack recall at 0.9886. The MLP reached an F1-score of 0.8807, while Logistic Regression produced the lowest F1-score at 0.8648. These results indicate that nonlinear tree-based ensemble models were the strongest methods in this experiment.

The XGBoost confusion matrix contained 6,607 true negatives, 2,381 false positives, 153 false negatives, and 10,859 true positives. Thus, only 153 of 11,012 attack records were missed, consistent with the 0.9861 attack recall. However, 2,381 of 8,988 normal records were classified as attacks, corresponding to a normal-class recall of 0.7351. The operational implication is important: the model is highly sensitive to attacks but would generate a substantial false-alert workload if its default classification threshold were deployed without calibration.

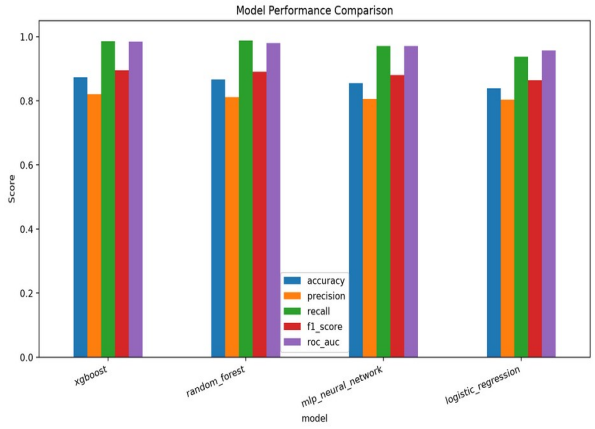



Fig. 1. Comparative performance of the four evaluated classifiers.

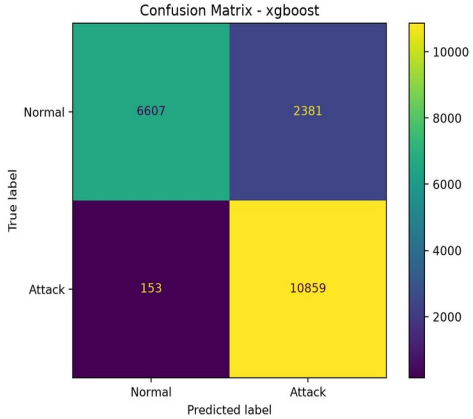



Fig. 2. Confusion matrix for the best model, XGBoost.

Efficiency results further favor XGBoost in this run. Its training time was approximately 1.124 s, close to Logistic Regression (1.039 s) and substantially lower than Random Forest (4.741 s) and the MLP (4.173 s). XGBoost inference took 0.112 s for the full 20,000-record test set. Because timing depends on hardware, library versions, parallelism, and system load, these values should not be generalized beyond the recorded environment; they are nevertheless useful for within-run comparison.

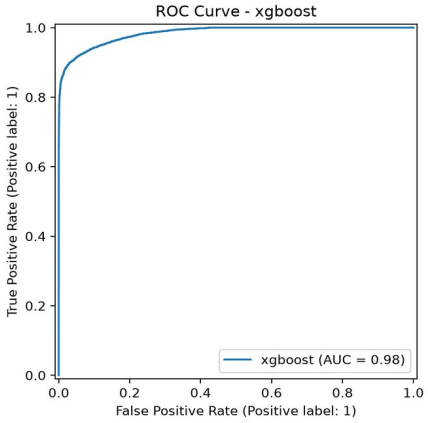



Fig. 3. ROC curve for XGBoost on the held-out test set.

## VII. Explainability and Decision Intelligence

### *A. Permutation Importance*

Permutation importance was computed for the selected XGBoost pipeline on a reproducible sample of up to 3,000 test records, using F1-score as the scoring function and three permutations per feature. The source time-to-live feature (sttl) was dominant, with a mean F1 decrease of 0.1998 when permuted. The next-ranked features were smean (0.0229), sbytes (0.0212), ct_dst_src_ltm (0.0183), and service (0.0134). The large gap between sttl and the remaining variables suggests that source TTL patterns carry substantial discriminatory information in this fitted model. Permutation importance reflects predictive dependence in this model and dataset; it should not be interpreted as causal evidence.

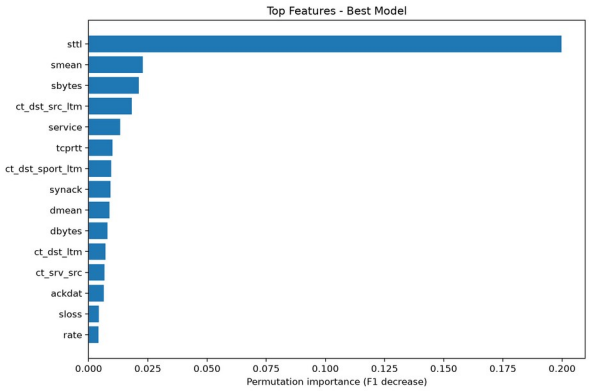



Fig. 4. Top permutation-important features for the selected XGBoost pipeline.

*B. Risk-Aware Decision Output*

The XGBoost attack probability was mapped to the four risk levels defined in Section IV. Across the 20,000 test records, 5,474 were assigned Low risk, 1,286 Moderate, 1,681 High, and 11,559 Critical. The observed attack prevalence within these bands was 0.46% for Low, 9.95% for Moderate, 23.50% for High, and 90.53% for Critical. This monotonic increase is directionally consistent with the intended use of model probability as a prioritization signal, although the High band contains many normal records and the thresholds were not optimized for calibration or operational cost.

The decision layer maps Low risk to continued monitoring; Moderate risk to increased logging and inspection of suspicious endpoints; High risk to escalation and isolation of suspicious traffic; and Critical risk to immediate incident-response review and containment procedures. This mapping demonstrates the strategic decision-intelligence concept: model outputs become prioritized actions rather than remaining isolated probability scores. In a production security operations center, these recommendations should be combined with asset criticality, threat intelligence, analyst judgment, and organizational response policy.

## VIII. Discussion

Three findings stand out. First, XGBoost offered the best overall predictive balance. Its F1-score exceeded Random Forest by approximately 0.0043, the MLP by 0.0148, and Logistic Regression by 0.0307. The difference is modest relative to Random Forest, but XGBoost also achieved the highest ROC-AUC and substantially shorter training time in this run. The result is consistent with the strength of gradient-boosted trees on heterogeneous tabular data [6], but it should not be interpreted as evidence that XGBoost universally outperforms other intrusion-detection methods.

Second, recall and false-positive burden must be considered together. Both XGBoost and Random Forest detected more than 98% of attack records, a desirable property when missed intrusions are costly. Yet XGBoost falsely flagged 2,381 normal records. In operational terms, optimizing only attack recall could overwhelm analysts with alerts. Threshold tuning, probability calibration, cost-sensitive learning, and context-aware filtering are therefore logical next steps.

Third, the experiment illustrates why decision intelligence should be evaluated separately from classification. The probability-to-risk transformation provides a usable prioritization mechanism, but the empirical attack rate of only 23.50% in the High band shows that intuitive probability bands do not automatically correspond to well-calibrated operational risk categories. Decision thresholds should ultimately be derived from validation data and explicit cost functions—for example, the relative costs of a missed intrusion, analyst review, service isolation, and business interruption.

The feature-importance analysis also offers a governance benefit. Explainability literature argues that interpretation can improve transparency and support appropriate human oversight of complex models [7], [8]. Here, the strong importance of sttl and smaller contributions from packet-size, byte-volume, connection-context, service, and TCP-timing features give analysts a starting point for examining why the model separates attack and normal traffic. Further local explanations would be needed to justify individual alerts.

## IX. Limitations and Threats to Validity

The study has several limitations. First, it uses a 50,000/20,000 subset rather than the full official UNSW-NB15 partitions, so results are specific to the prepared sample. Second, the evaluation is binary; performance across individual attack families is not optimized or formally assessed as a multiclass task. Third, only one train/test run with fixed seed 42 is reported, without repeated cross-validation, confidence intervals, or significance tests. Fourth, hyperparameters were predefined rather than selected through a systematic tuning protocol. Fifth, the benchmark was generated in a controlled cyber-range environment and may not reflect contemporary production traffic or current attack distributions. Sixth, the decision thresholds are heuristic and were not optimized for calibration, expected cost, analyst workload, or organization-specific risk appetite. Finally, timing measurements are machine-dependent and should be interpreted only as within-run comparisons.

## X. Conclusion and Future Work

This study implemented and evaluated an end-to-end local pipeline for AI-driven cybersecurity threat detection and strategic decision intelligence. On the prepared UNSW-NB15 subset, XGBoost was the strongest model by F1-score, reaching 0.8955 F1, 0.9861 attack recall, and 0.9841 ROC-AUC while maintaining a short training time. Random Forest achieved slightly higher recall but lower overall F1 and ROC-AUC. The MLP did not outperform the tree ensembles, and Logistic Regression served as a useful linear baseline. Permutation importance identified sttl as the most influential feature in the selected model. Mapping predicted attack probabilities into risk levels and recommended actions demonstrated a practical bridge from predictive analytics to decision support.

Future work should evaluate the full official UNSW-NB15 partitions, repeat experiments across multiple seeds, tune models on a validation set, report confidence intervals, and perform multiclass attack-family detection. Probability calibration and cost-sensitive threshold optimization are particularly important before the decision-intelligence layer is interpreted operationally. Additional work should compare SHAP or other local explanation methods, test more recent intrusion datasets, and evaluate the framework with security analysts in realistic workflows. The broader objective is not merely to maximize classification accuracy, but to produce security intelligence that is accurate, explainable, prioritized, and usable in human-governed cyber defense.

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